

Review: Solving one-step inequalities



1 Determine whether the following values satisfy the inequality $x + 3 > 5$.

a $x = 1$

b $x = 3$

c $x = -9$

d $x = 2$

2 Solve the following inequalities.

a $x + 5 \geq 11$

b $9 > y + 4$

c $n + 8 \leq 7$

d $m - 5 > 9$

e $7 \leq x - 3$

f $x - 8 < 11$

g $x - 8 \leq -4$

h $9y > 81$

i $10x < 90$

j $56 \leq 8p$

k $3x \leq -18$

l $\frac{x}{3} \geq 6$

m $\frac{n}{7} > 8$

n $8 \geq \frac{x}{3}$

o $-4x < 32$

p $-4y \geq -16$

q $-5x \leq -40$

r $-24 < -8m$

s $\frac{a}{6} < 11$

t $\frac{x}{-9} \geq 5$

u $\frac{x}{-7} < 2$

3 For each of the following:

i Write down the inequality described by the statement.

ii Solve for the value of x .

a "Seven more than the value of x is at least nine".

b "Half of x is no more than four".

c "Eight is greater than the result of taking seven away from x ".

d "Negative eight groups of x is less than three".

4 For each of the following:

i Solve the inequality.

ii Plot the solution to the inequality on a number line.

a $8 + x \leq 8$

b $x + 1 < 4$

c $x - 4 < 1$

d $3 + x < 2$

e $-8 < x - 4$

f $-2 + x \leq 4$

g $2 + x \geq 7$

h $8 < 10 + x$

i $-5 + x < -7$

j $x - 2 \geq -3$

k $-4 < -3 + x$

l $6x \leq 48$

$$\text{m } 36 \leq 9x$$

$$\text{o } 7x < -35$$

$$\text{q } -3x > -21$$

$$\text{s } \frac{x}{2} \geq 7$$

$$\text{u } -6 \geq \frac{x}{3}$$

$$\text{w } \frac{x}{-5} > 2$$

$$\text{y } -2 \leq \frac{x}{-7}$$



$$2x > -4$$

$$\text{p } -5x \leq 15$$

$$\text{r } 72 > -8x$$

$$\text{t } \frac{x}{6} < -2$$

$$\text{v } \frac{x}{-8} \leq -2$$

$$\text{x } \frac{x}{-7} < 2$$

$$\text{z } \frac{x}{3} > \frac{4}{3}$$

5 For each of the following:

i Solve the inequality.

ii Plot the solution to the inequality on a number line.

$$\text{a } x + \frac{7}{4} \geq \frac{9}{4}$$

$$\text{b } -4.5x \geq -18$$

$$\text{c } x + 2.5 \leq 7.5$$

$$\text{d } x + 3.5 > 8$$

$$\text{e } x - 2.6 \leq 2.4$$

$$\text{f } -4.5 + x \geq 4$$